(Sparse) Attention to the Details: Preserving Spectral Fidelity in ML-based Weather Forecasting Models

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Abstract

We introduce MOSAIC, a probabilistic weather forecasting model that addresses two principal sources of spectral degradation in ML-based weather prediction: (1) deterministic training against ensemble means and (2) compressive encoding creating an information bottleneck. MOSAIC generates ensemble members through learned functional perturbations and operates on native-resolution grids via block-sparse attention, a hardware-aligned mechanism that captures long-range dependencies at linear cost by sharing keys and values across spatially adjacent queries. At 1.5° resolution with 214M parameters, MOSAIC matches or outperforms models trained on 6× finer data on headline upper-air variables and achieves state-of-the-art results among 1.5° models, producing well-calibrated ensembles whose individual members exhibit near-perfect spectral alignment across all resolved frequencies. A 24-member, 10-day forecast takes under 12 s on a single H100 GPU.

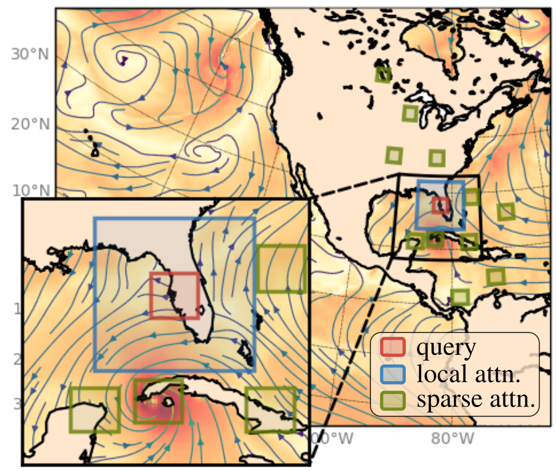

Figure 1. Block-sparse attention for weather forecasting. Spatially close query tokens (red block over Tampa Bay) collectively attend to both local key-value pairs (blue block over Florida) and dynamically selected, spatially distributed ones (green blocks). Sparse attention enables capturing long-range dependencies in high-resolution weather data, critical for extreme events such as hurricane formation (note the eye visible in the inset).

1. Introduction

Accurate weather forecasts save lives and enable timely decisions during extreme events. Phenomena such as frontal zones and tropical cyclones cause catastrophic damage and span relatively short distances, requiring models that faithfully resolve fine spatial scales. Numerical weather prediction (NWP) systems achieve this by integrating the equations of fluid dynamics, thermodynamics, and radiative transfer, but at a computational cost that scales cubically with grid resolution.

ML-based weather prediction models (MLWPs; Bi et al. 2023; Lam et al. 2023; Bodnar et al. 2025) have emerged as efficient surrogates, generating 10-day forecasts in under 60 seconds on a single GPU, achieving a 1000-10000× speedup over NWP (Buizza et al., 2018; Bauer et al., 2020). However, these models struggle to reproduce the power spectra of atmospheric variables, systematically suppressing spectral power at fine scales (Bonavita, 2024). This happens for two main reasons. First, most MLWPs are deterministic and are trained to approximate the conditional mean over future states, which is inherently smoother than any individual realization. Probabilistic MLWPs (Price et al., 2023; Lang et al., 2024; Alet et al., 2025) address this limitation by producing ensemble members rather than a single mean prediction. Each member represents a plausible realization that can exhibit the sharp, fine-scale structures present in nature with spectral properties substantially closer to ground truth than their deterministic counterparts.

Second, most MLWPs use compressive encoding in their model architecture: they compress high-resolution weather data to a coarser mesh before processing, with a coarsening factor that typically far exceeds the increase in the number of feature channels, creating an information bottleneck shown to cause irreversible information loss in both

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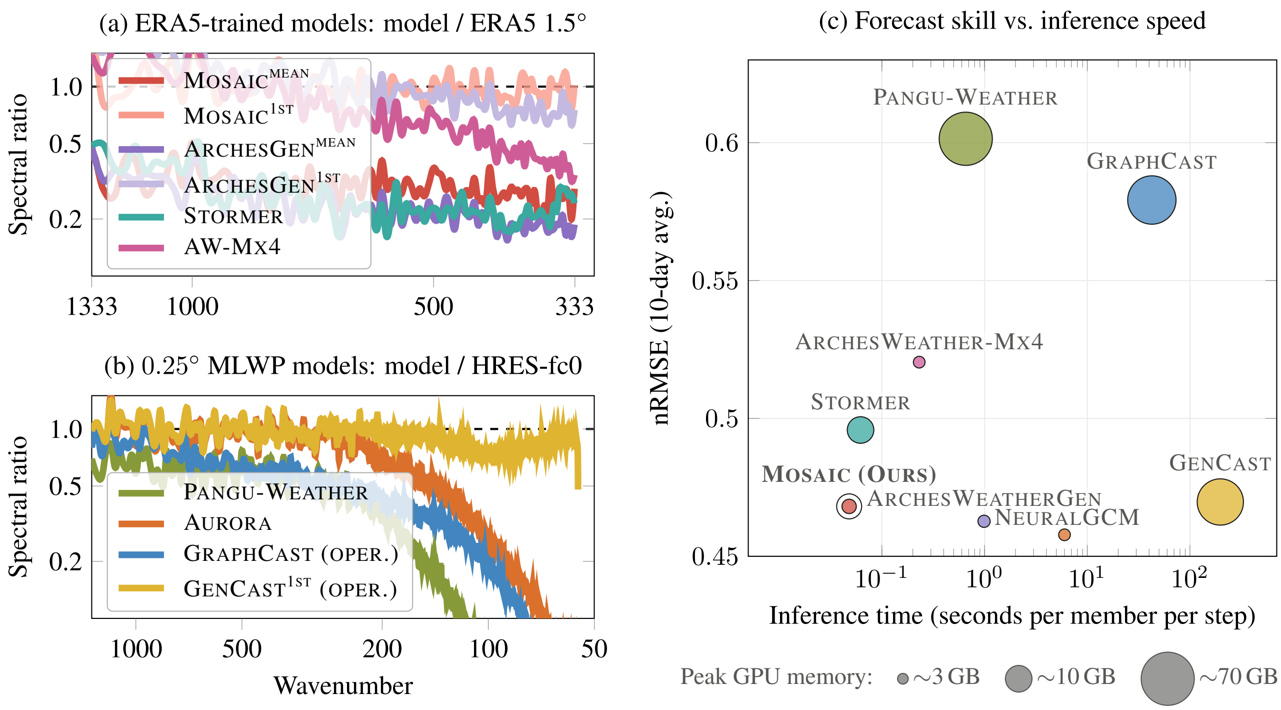

Figure 2. Spectral analysis and efficiency of MLWP models. **(a, b)** Spectral power ratios (model / reference) of 10-meter wind speed from a single 10-day forecast at 1.5° (ERA5) **(a)** and 0.25° (HRES-fc0) **(b)** resolution. Individual ensemble members of probabilistic models (MOSAIC, ARCHESGEN, GENCAST) demonstrate stronger spectral coherence with the reference, while ensemble means and deterministic models (STORMER, ARCHESWEATHER-MX4, GRAPHCAST, AURORA, PANGU-WEATHER) deviate substantially, with divergence most pronounced at fine spatial scales. **(c)** Forecast skill vs. inference speed on a single H100 GPU; marker size indicates peak GPU memory. nRMSE is computed at 240 h lead time at 1.5° resolution over a set of key variables (see Appendix A.8 for details).

vision (Wang et al., 2025) and weather models (Nguyen et al., 2023). Although compression-based approaches are efficient, they increase the complexity of the learned function: fine-scale spatial variations originally distributed across neighboring mesh nodes become entangled within channel dimensions of coarse nodes, requiring the model to jointly recover geometric structure and spectral content.

In this work, we introduce MOSAIC: a probabilistic weather forecasting model designed to address both sources of spectral degradation. First, we follow (Alet et al., 2025) and use learned functional perturbations to incorporate uncertainty, producing ensemble forecasts whose individual members preserve realistic spectral variability. Second, we propose block-sparse attention: a hardware-aligned formulation of native sparse attention (Yuan et al., 2025) that exploits the intrinsic locality of physical data by sharing keys and values across spatially adjacent queries, computing block-to-block interactions. Each block dynamically selects which regions of the grid to attend to, capturing arbitrarily long-range dependencies at linear cost. To ensure efficient memory access, we process data on the HEALPix mesh (Gorski et al., 1999), which offers contiguous storage and therefore allows us to operate on blocks. Together, these design choices make MOSAIC directly applicable to high-resolution grids without compression.

The main contributions of this work are:

- We propose block-sparse attention, a hardware-aligned sparse attention mechanism that captures long-range dependencies at linear cost, enabling weather models to operate on native-resolution grids without compressive encoding.
- We introduce MOSAIC, a probabilistic weather forecasting model that uses block-sparse attention and learned functional perturbations to address both the architectural and statistical sources of spectral degradation in MLWPs.
- We demonstrate that MOSAIC at 1.5° resolution matches or outperforms models trained on 6× finer data on headline upper-air variables, produces well-calibrated ensembles with near-perfect spectral alignment, and generates a 24-member, 10-day forecast in under 12 s on a single H100 GPU.

2. Related Work

2.1. Effective Resolution in Weather Forecasting

Abdalla et al. (2013) define the effective resolution of a weather prediction model as the smallest spatial scale it fully resolves – the wavelength at which the model's power spectrum starts to deviate from the ground truth. Multiple studies (Bonavita, 2024; Husain et al., 2025) show that MLWPs fail to reproduce realistic spectra and therefore exhibit low effective resolution, struggling to resolve sharp phenomena such as frontal zones and tropical cyclones that span 50-80 km, despite being trained on data at 28 km spatial resolution. Gupta et al. (2025) find systematic underestimation of spectral power at mesoscales (10-100 km) across multiple MLWPs. Li et al. (2025) demonstrate that Pangu (Bi et al., 2023) underestimates kinetic energy at wavelengths below 1000 km and fails to replicate the characteristic $-\frac{5}{3}$ spectral slope of physics-based models.

Several strategies address this spectral degradation. Subich et al. (2025) modify the training objective to penalize spectral discrepancy directly, improving GraphCast's effective resolution from 1,250 to 160 km. Similarly, Bonev et al. (2025) also optimize in the spectral domain, achieving realistic spectra at subseasonal lead times. Hybrid approaches (Kochkov et al., 2023; Husain et al., 2025) combine ML with physics-based solvers, leveraging the numerical backbone for fine-scale structure. Post-hoc methods (Lippe et al., 2023; Oommen et al., 2024) condition diffusion models on smooth predictions to recover high-frequency content. Most directly related to our work, Baño-Medina et al. (2025); Nordhagen et al. (2025) process on the original high-resolution mesh with message-passing neural networks and observe significantly better spectral correspondence. We follow the same principle, but replace the fixed graphs with sparse attention, which dynamically determines interactions based on the current weather state.

2.2. Compression Effect on Expressivity

The effect of compressive encoding on model expressivity is well studied in computer vision, where patchification, the core tokenization strategy of vision transformers (Dosovitskiy et al., 2020), reduces computational cost by compressing spatial information. (Wang et al., 2025) demonstrate irreversible information loss caused by patchification and show that test loss declines consistently as patch size decreases, reaching optimal performance at 1×1 patches. Moreover, reducing patch size is more beneficial than increasing model parameters, indicating that added capacity cannot compensate for compression-induced information loss. The same pattern holds in weather forecasting, where Nguyen et al. (2023) show that decreasing patch size consistently improves forecast accuracy.

For atmospheric data, where fine-scale spatial structure is associated with extreme weather events, this information loss is especially consequential. We avoid compressive encoding entirely: rather than projecting high-resolution data onto a coarse mesh, we operate directly at native resolution via block-sparse attention to preserve details.

2.3. Ultra-scale Grid Processing

Avoiding compression shifts the bottleneck to efficiently processing the original ultra-scale data. Standard attention (Vaswani et al., 2017) scales quadratically with sequence length; FlashAttention (Dao et al., 2022) reduces the memory cost to linear, but still has the quadratic complexity. Linear attention (Yang et al., 2024; 2023) achieves linear cost by replacing the softmax with a kernel-based approximation that maintains a fixed-size state, but sacrifices the input-dependent selectivity that makes standard attention expressive. Native Sparse Attention (NSA; Yuan et al. 2025) resolves this trade-off by computing the dot product over a dynamically selected subset of key-value pairs per query, retaining softmax expressivity at linear cost.

In scientific computing, the need to process large-scale physical simulations has led to complementary approaches. Holzschuh et al. (2025) handle ultra-scale grids (256^3) by partitioning the domain into non-overlapping subdomains processed in parallel. Alkin et al. (2024); Wu et al. (2024) use learnable pooling to compress the input into a coarser representation where the bulk of computation occurs. Zhdanov et al. (2025); Brita et al. (2025) avoid pooling and instead use hierarchical trees to impose structure on irregular data, enabling sparse attention with linear cost.

We adapt sparse attention for high-resolution weather grids but further exploit spatial locality: rather than selecting key-value blocks independently per query, we group spatially proximate queries into blocks that jointly select which regions to attend to. This block-level sparsity aligns well with GPU memory access patterns, enabling efficient processing of grids with hundreds of thousands of points without domain decomposition or lossy pooling.

3. Theoretical Background

3.1. Problem Formulation

We frame weather forecasting as sampling from the conditional distribution $p\left(X^t \mid X^{t-1}, \ldots, X^{t-H}\right)$ over the next atmospheric state X^t given the history of previous states, where each state consists of both surface- and pressure-level atmospheric variables on a latitude-longitude grid. A T-step ensemble forecast is generated by sampling autoregressively:

$$X^t \sim p\left(X^t \mid X^{t-1}, \ldots, X^{t-H}\right), \quad t = 1, \ldots, T. \quad (1)$$

Drawing multiple trajectories from this process yields an ensemble that quantifies forecast uncertainty.

3.2. Hierarchical Sphere Tessellation

HEALPix (Gorski et al., 1999) is a hierarchical, equal-area tessellation of a sphere into four-sided polygons (pixels). The sphere is partitioned into a tree-like hierar-

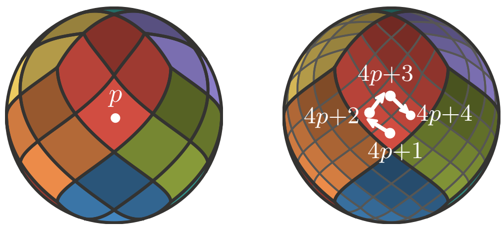

Figure 3. HEALPix mesh refinement. Each pixel (left) is subdivided into four children (right), whose indices follow a Z-order curve that keeps spatially close pixels contiguous in memory.

chy starting from 12 base pixels (4 around each pole, 4 around the equator), with each pixel recursively subdivided into four children, see Fig. 3. At a given resolution, each pixel covers an identical surface area, ensuring that signal sampling and noise integration are geographically unbiased, unlike traditional latitude-longitude grids, which over-sample near the poles.

The size parameter N_{side} defines a HEALPix mesh and determines both the total number of pixels $12N_{\text{side}}^2$ and the approximate angular resolution:

N_{side}	32	64	128	256
# pixels	12,288	49,152	196,608	786,432
resolution	1.83°	0.92°	0.46°	0.23°

The pixels are organized in memory such that spatially close regions occupy consecutive indices[1]. A pixel with index p at resolution N_{side} subdivides into four children with consecutive indices $4p, 4p{+}1, 4p{+}2, 4p{+}3$ at resolution $2N_{\text{side}}$. This contiguous, hierarchical layout makes HEALPix particularly suited for block-based computation: loading a spatially local block of pixels requires a single coalesced memory read rather than the scattered accesses needed on a latitude-longitude grid. Several ML-based weather models already exploit this structure (Ramavajjala, 2024; Linander et al., 2025; Karlbauer et al., 2023).

3.3. Native Sparse Attention

Native Sparse Attention (NSA; Yuan et al. (2025)) addresses the quadratic complexity of standard self-attention by restricting each query's interactions to a dynamically determined subset of keys and values. The computation is organized into three branches – compression, selection and local – whose outputs $\mathbf{o}_i^{CG}$ (coarse-grained), $\mathbf{o}_i^{FG}$ (fine-grained) and $\mathbf{o}_i^{L}$ (local) are combined via learned gating.

$$\mathbf{o}_i = g^{CG}(\mathbf{x}_i)\cdot\mathbf{o}_i^{CG} + g^{FG}(\mathbf{x}_i)\cdot\mathbf{o}_i^{FG} + g^{L}(\mathbf{x}_i)\cdot\mathbf{o}_i^{L} \quad (2)$$

where g^{CG}, g^{FG}, g^{L} are learnable linear gating functions.

[1]This corresponds to the NESTED ordering scheme. We refer the reader to (Gorski et al., 1999) for information about the alternative RING scheme.

Compression Let $\{B_1, \ldots, B_m\}$ be a partition of tokens into m non-overlapping blocks. For each B_j, let $\mathbf{K}_j = \{\mathbf{k}_l \mid l{\in}B_j\}$ denote the set of keys in the block. A block representation is computed via a learnable function φ:

$$\bar{\mathbf{k}}_j = \varphi(\mathbf{K}_j). \quad (3)$$

Coarse-grained values are obtained as $\bar{\mathbf{v}}_j = \varphi(\mathbf{V}_j)$ with $\mathbf{V}_j = \{\mathbf{v}_l \mid l \in B_j\}$. Coarse-grained attention between query i and each block is then evaluated, with attention scores retained for the selection branch:

$$a_{ij} = \frac{\exp\left(\mathbf{q}_i^\top \bar{\mathbf{k}}_j/\sqrt{d_k}\right)}{\sum_{l=1}^{m}\exp\left(\mathbf{q}_i^\top \bar{\mathbf{k}}_l/\sqrt{d_k}\right)}, \quad \mathbf{o}_i^{CG} = \sum_{j=1}^{m} a_{ij}\bar{\mathbf{v}}_j. \quad (4)$$

The compression branch both captures global context and guides sparsification in the selection branch.

Selection For each query i, NSA selects the top-n blocks with the highest coarse-grained attention scores, $\mathcal{S}_i = \text{top-}n\,(a_{i,:})$, and computes fine-grained attention over keys and values *within* the selected blocks:

$$\mathbf{o}_i^{FG} = \sum_{j\in\mathcal{S}_i}\sum_{l\in B_j}\frac{\exp\left(\mathbf{q}_i^\top \mathbf{k}_l/\sqrt{d_k}\right)}{Z_i}\mathbf{v}_l \quad (5)$$

where $Z_i = \sum_{j\in\mathcal{S}_i}\sum_{l\in B_j}\exp\left(\mathbf{q}_i^\top \mathbf{k}_l/\sqrt{d_k}\right)$ is the normalizing constant. By operating at full resolution, the selection branch preserves fine-scale detail while capturing long-range interactions.

Local Attention The local branch applies standard attention for each query i over keys and values within a sliding window, yielding $\mathbf{o}_i^{L}$. This frees the compression and selection branches to focus on long-range interactions.

4. MOSAIC

Tobler's first law of geography states that "nearby things are more related than distant things" (Tobler, 1970). This principle motivates two important design choices in MOSAIC: (1) representing data on the HEALPix mesh, where spatially close points occupy contiguous memory and (2) grouping spatially adjacent tokens into blocks that jointly determine which regions of the globe to attend to.

4.1. Data Representation on HEALPix

Weather forecast data is traditionally stored on latitude-longitude grids, where points are ordered row-wise along parallels. This layout places spatially close points far apart in index space, requiring scattered memory accesses to load a block of neighboring pixels. We instead operate on the HEALPix mesh, which guarantees that spatial neighbors occupy contiguous memory locations, enabling coalesced GPU reads and hardware-aligned block computation.

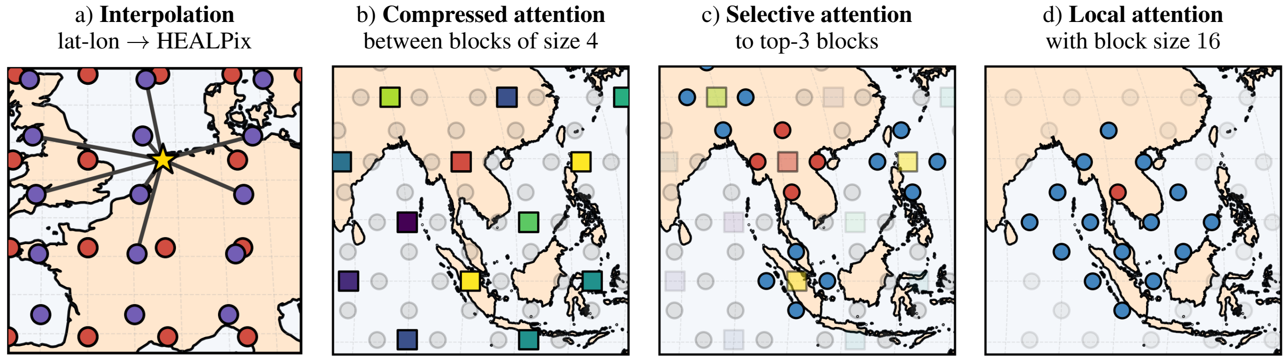

Figure 4. Block-sparse attention for weather forecasting. **(a)** Weather data is interpolated from a latitude-longitude grid (red) to the HEALPix mesh (purple) via cross-attention. **(b–d)** The three branches of block-sparse attention, illustrated for a single query block (red): **(b)** Compression computes attention between coarse-grained block representations (squares; color indicates attention score). **(c)** Selection attends at full resolution to the top-n key blocks. **(d)** Local attention captures fine-grained interactions within local blocks.

Interpolating Between Grids To transfer data from the latitude-longitude grid to the HEALPix mesh, we employ the cross-attention interpolation scheme of (Wessels et al., 2025). For each target point i on the HEALPix mesh and neighboring source point j on the latitude-longitude grid, queries are computed from the relative position $\mathbf{p}_{ij}$, while keys and values are derived from source features:

$$\mathbf{q}_{ij} = W_q \frac{\mathbf{p}_{ij}}{\|\mathbf{p}_{ij}\|}, \quad \mathbf{k}_j = W_k \mathbf{x}_j, \quad \mathbf{v}_j = W_v \mathbf{x}_j \tag{6}$$

The interpolated output at target position i is:

$$\mathbf{o}_i = \sum_{j \in \mathcal{N}_i} \text{softmax}_j \left(\mathbf{q}_{ij}^\top \mathbf{k}_j / \sqrt{d} \right) \mathbf{v}_j \tag{7}$$

where $\mathcal{N}_i$ denotes the set of source grid points neighboring i.

4.2. Block-Sparse Attention

BSA retains the three-branch structure of NSA – compression, selection, and local – combined via learned gating (Eq. 2). The key difference is that sparsity operates at the block level: rather than each query token independently selecting which key blocks to attend to, tokens within a query block jointly select key blocks. This reduces compression from token-to-block to block-to-block interactions and amortizes the selection cost across all tokens in a block.

Compression We partition N tokens into m non-overlapping blocks $\{B_1, \ldots, B_m\}$ and compute coarse-grained representations of queries, keys, and values via a function φ (mean pooling in our experiments):

$$\bar{\mathbf{q}}_i = \varphi(\mathbf{Q}_i), \quad \bar{\mathbf{k}}_j = \varphi(\mathbf{K}_j), \quad \bar{\mathbf{v}}_j = \varphi(\mathbf{V}_j). \tag{8}$$

Attention is computed at the block level between query block i and key block j:

$$\bar{a}_{ij} = \frac{\exp\left(\bar{\mathbf{q}}_i^\top \bar{\mathbf{k}}_j / \sqrt{d_k}\right)}{\sum_{l=1}^{m} \exp\left(\bar{\mathbf{q}}_i^\top \bar{\mathbf{k}}_l / \sqrt{d_k}\right)} \tag{9}$$

The coarse-grained output is then distributed to all tokens within each query block:

$$\bar{\mathbf{o}}_i^{CG} = \sum_{j=1}^{m} \bar{a}_{ij} \bar{\mathbf{v}}_j, \qquad \mathbf{o}_l^{CG} = \bar{\mathbf{o}}_i^{CG} \quad \forall l \in B_i. \tag{10}$$

Coarse-grained attention captures broad spatial patterns in a computationally feasible manner. For instance, a block over the Netherlands might attend strongly to blocks over the North Atlantic or the Arctic, identifying synoptic-scale influences before the selection branch zooms in on details.

Selection For each query block i, the top-n key blocks with highest coarse-grained attention scores are selected, $\bar{\mathcal{S}}_i = \text{top-}n\left(\bar{a}_{i,:}\right)$. Fine-grained attention is then computed between all tokens in query block i and all tokens in the selected key blocks, capturing long-range dependencies:

$$\mathbf{o}_l^{FG} = \sum_{j \in \bar{\mathcal{S}}_i} \sum_{t \in B_j} \frac{\exp\left(\mathbf{q}_l^\top \mathbf{k}_t / \sqrt{d_k}\right)}{Z_l} \mathbf{v}_t \quad \forall l \in B_i \tag{11}$$

where $Z_l = \sum_{j \in \bar{\mathcal{S}}_i} \sum_{t \in B_j} \exp\left(\mathbf{q}_l^\top \mathbf{k}_t / \sqrt{d_k}\right)$ is the normalizing constant. The selection branch resolves fine-scale structure within the chosen regions, for example, by capturing how a specific frontal zone over the Atlantic influences local conditions in Western Europe.

Local Attention To capture fine-grained local interactions, we compute attention within large blocks of pixels independently. Unlike NSA's sliding window, which would require handling irregular boundaries on the sphere, block attention aligns naturally with the HEALPix structure and can be computed in parallel. The local branch resolves fine-grained structure within each region, such as temperature gradients across a coastline or wind patterns within a valley, freeing other branches to focus on long-range interactions.

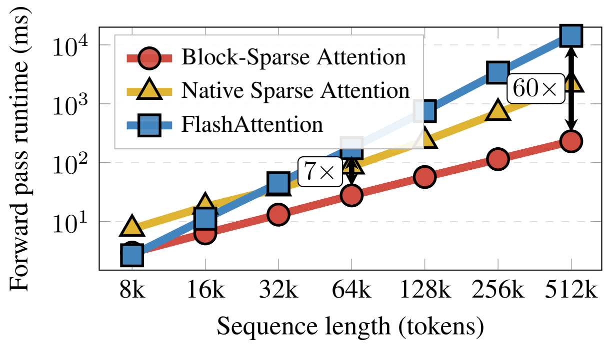

Figure 5. Forward pass runtime vs. sequence length for Block-Sparse Attention, NSA, and full FlashAttention; measured on NVIDIA RTX A4500. See Appendix A.3 for details.

Computational cost Let b denote the block size yielding $\frac{N}{b}$ blocks in total. The combined cost of block-sparse attention across branches is: $\mathcal{O}\big(N^2/b^2+Nnb+Nb\big)$. The compression branch (CG) computes all pairwise interactions between coarse-grained blocks resulting in $\mathcal{O}(N^2/b^2)$ cost. For block sizes $b \geq 128$, this cost does not form a bottleneck. The selection branch (FG) computes, for each token, attention over n selected blocks of size b yielding $\mathcal{O}(Nnb)$ cost, which is linear in N as n and b are fixed hyperparameters. Local attention is $\mathcal{O}(Nb)$ since each token only attends to tokens within the same block. Together, these costs make block-sparse attention scalable to weather grids with hundreds of thousands of tokens. In practice, BSA scales near-linearly with sequence length, achieving up to 61.8× speedup over dense attention and 9.4× over NSA (Fig. 5). We implement BSA in Triton (Tillet et al., 2019) following the memory-efficient approach of FlashAttention (Dao et al., 2022); see Appendix A.3 for details.

4.3. Model Architecture

Weather dynamics spans a wide range of spatial scales, from planetary waves spanning thousands of kilometers to mesoscale convective systems tens of kilometers across. To capture this multi-scale structure, we adopt a U-Net architecture that processes data at progressively coarser resolutions in the encoder path, then refines predictions back to the original resolution in the decoder path. Each resolution level consists of block-sparse attention layers that capture interactions at that scale. Skip connections between corresponding encoder and decoder levels preserve fine-grained information that might otherwise be lost during coarsening. Crucially, the first encoder stage computes spatial interactions via block-sparse attention at native resolution before any coarsening occurs. This distinguishes MOSAIC from compression-based MLWPs, which first reduce spatial resolution (e.g., via patchification) and then compute interactions on the compressed representation, creating an information bottleneck at the finest scales.

Coarsening The encoder path progressively coarsens the HEALPix mesh. At each coarsening step, a parent pixel at resolution N_{side} aggregates features from its four children at resolution $2N_{\text{side}}$ via learnable pooling:

$$\mathbf{x}_{\text{parent}} = W_x^{\downarrow}\mathbf{X}_c + W_p^{\downarrow}\Delta\mathbf{P}_c, \tag{12}$$

where $\mathbf{X}_c = [\mathbf{x}_{c_i}]_{i=1}^4$ and $\Delta\mathbf{P}_c = [\Delta\mathbf{p}_{c_i}]_{i=1}^4$ are the stacked child features and relative positions, with $\Delta\mathbf{p}_{c_i} = \mathbf{p}_{c_i} - \mathbf{p}_{\text{parent}}$. Here $W_x^{\downarrow}, W_p^{\downarrow}$ are learnable projections. At the coarsest resolution, a bottleneck stage applies additional transformer blocks before the decoder path begins.

Refinement The decoder path progressively refines features back to the original resolution. At each refinement step, a parent pixel at resolution N_{side} predicts features for its four children at resolution $2N_{\text{side}}$:

$$\mathbf{X}_c = W_x^{\uparrow}\mathbf{x}_{\text{parent}} + W_p^{\uparrow}\Delta\mathbf{P}_c. \tag{13}$$

The corresponding encoder features are then added to the refined features via skip connections.

Overall architecture Input features that include both dynamic (e.g. 2-meter temperature, 10-meter wind components) and static variables (e.g. land-sea mask, soil type) are combined with sinusoidal time (day and year) encodings (Bodnar et al., 2025) and projected to the hidden dimension via a two-layer MLP followed by interpolation to the HEALPix mesh (see Eq. 7).

The data then passes through the U-Net, where a sequence of transformer blocks is applied at each resolution level. Each block has a standard pre-norm structure (Touvron et al., 2023) with block-sparse attention and SwiGLU (Shazeer, 2020) feed-forward network:

$$\begin{aligned} \mathbf{x} &\leftarrow \mathbf{x} + \text{Attention}(\text{RMSNorm}(\mathbf{x})) \\ \mathbf{x} &\leftarrow \mathbf{x} + \text{FFN}(\text{RMSNorm}(\mathbf{x}), \mathbf{z}) \end{aligned} \tag{14}$$

We use grouped query attention (GQA; (Ainslie et al., 2023)) to reduce memory bandwidth requirements, with queries attending to a number of shared key-value heads. We apply 2D rotary positional embeddings (RoPE; (Heo et al., 2024)) to queries and keys, encoding longitude and latitude positions. The head dimension is split equally between the two coordinates, with each half receiving standard RoPE encoding for its respective angle in radians.

After the decoder, features are interpolated back to the latitude-longitude grid and projected to the output dynamic variables to predict the next weather state.

4.4. Uncertainty Quantification

Block-sparse attention addresses the architectural source of spectral degradation by avoiding compression. To address the statistical source, MOSAIC produces ensemble forecasts via learned functional perturbations (Alet et al., 2025), a form of Bayesian neural networks (Blundell et al., 2015). The key idea is to inject noise into the parameters of the model, resulting in a single weight perturbation affecting the entire forecast in a globally consistent way. Alet et al. (2025) condition parameters of layer normalization layers on the noise vector; we instead inject noise into SwiGLU layers as bias added inside the gate:

$$\text{cSwiGLU}(x, z) = (\sigma(xW_g + z) \odot (xW_v))\, W_{out} \tag{15}$$

where $\odot$ denotes element-wise multiplication, and $\sigma(x) = \frac{x}{1+e^{-x}}$ is the Swish activation. We found simple additive noise injection in the SwiGLU gate to work best. Mathematically, this induces an effective output projection $W_{out}^{\text{eff}} = \text{diag}(s) \cdot W_{out}$, where scaling factors s_i are drawn from input-dependent distributions determined by the gate activations xW_g. The learned weights W_{out} define the mean structure, while noise z flowing through the input-dependent nonlinearity defines a learned, adaptive covariance. This produces structured uncertainty as the model learns which features should vary across ensemble members based on the input and a small number of latent variables, rather than applying uniform noise everywhere.

Training with CRPS The probabilistic model is trained by optimizing the Continuous Ranked Probability Score (CRPS; Alet et al. (2025)), a proper scoring rule for univariate distributions that encourages both accurate and well-calibrated probabilistic forecasts. We use an unbiased (fair) CRPS estimator (Zamo & Naveau, 2018), which for a single variable $y \in \mathbb{R}$ and an ensemble $x^{1:N}$ is given by

$$\begin{aligned}\text{CRPS}(x^{1:N}, y) := \frac{1}{N}\sum_n |x^n - y| \\ - \frac{1}{2N(N-1)}\sum_{n,n'} |x^n - x^{n'}|.\end{aligned} \tag{16}$$

The CRPS decomposes into a reliability term penalizing deviation from the ground truth and a sharpness term encouraging ensemble spread only where warranted by true uncertainty. The training loss is the latitude-weighted, variable-weighted fair CRPS averaged over all grid points:

$$\mathcal{L} := \frac{1}{|D|}\sum_d \frac{1}{HW}\sum_{h,w}\sum_{i=1}^{C} \alpha_i\, \omega_h\, \text{CRPS}(\hat{x}^{1:N}_{i,h,w,d}, \hat{y}_{i,h,w,d}) \tag{17}$$

where d indexes the batch, (h, w) indexes spatial grid points, i indexes the C=82 output channels, α_i is a per-channel variable weight (Lam et al., 2023), and ω_h is a latitude weight proportional to $\cos(\phi_h)$. Both predictions $\hat{x}$ and targets $\hat{y}$ are in standardized space. Full details are in Appendix A.5.

5. Experiments

We aim to answer the following research questions:

- **RQ1:** How does MOSAIC compare against SOTA MLWPs in terms of forecasting skill?
- **RQ2:** Does MOSAIC preserve spectral fidelity across all resolved frequencies?
- **RQ3:** Are its ensemble forecasts well-calibrated?
- **RQ4:** Does MOSAIC's spectral fidelity translate into skillful forecasts of real-world extreme events?

We evaluate MOSAIC under two benchmarks that differ in training data, step size, and test year: a controlled 1.5° comparison against other 1.5° models, and a cross-resolution comparison against state-of-the-art 0.25° MLWPs. We further assess extreme-event forecasting through a case study of Hurricane Ian. We also conduct an ablation study (Appendix B.2) where we modify a single aspect of the model relative to a controlled baseline.

Data All experiments share the same pretraining data: ERA5 reanalysis from 1979 to 2018 at 1.5° spatial resolution (121×240 equiangular grid). The difference comes down to the finetuning stage:

- **1.5° benchmark.** Following the protocol of Couairon et al. (2024), we finetune on ERA5 2007–2019 to account for distribution shift in the earlier reanalysis period, and test on 2020. Forecasts are evaluated against ERA5 ground truth. MOSAIC is trained with step size of 24 h to match the baselines.
- **0.25° benchmark.** We finetune on HRES-fc0 analysis from 2016 to 2021 and test on 2022. The evaluation is done by initializing from HRES-fc0, and evaluating against it. MOSAIC is trained with step size of 6 h as the majority of the baselines.

Baselines For the 1.5° benchmark, we compare against models operating at comparable resolution: Stormer (Nguyen et al., 2024), ArchesWeather-Mx4 (Couairon et al., 2024), ArchesWeatherGen (Couairon et al., 2024), NeuralGCM-ENS (Kochkov et al., 2023). We also include IFS HRES and IFS ENS (ECMWF) as NWP reference points. For the 0.25° benchmark, we compare against deterministic models (GraphCast (Lam et al., 2023), Aurora (Bodnar et al., 2025), Pangu-Weather (Bi et al., 2023)) and probabilistic models (FGN (Alet et al., 2025), GenCast (Price et al., 2023)), as well as IFS-ENS, IFS-HRES, and climatology. Training resources of all baselines are given in Table 2.

Training details MOSAIC contains 214M parameters organized in a U-Net architecture with 14 layers and hidden dimensions progressing from 768 (N_{side}=64) to 1024 (N_{side}=32) to 1280 (N_{side}=16). Full training is done with

	RES.	STEP	S/STEP[†]	Z500	T850	Q700	U850	V850	SP
IFS HRES	0.1°	1H	—	802.75	3.654	1848.0	6.460	6.530	748.68
KEISLER (2022)	1°	6H	—	787.01	3.559	1659.5	6.106	6.121	N/A
STORMER ENS (MEAN)	1.4°	24H	0.06	665.88	3.001	1445.4	5.198	5.137	619.89
ARCHESWEATHER-MX4	1.5°	24H	0.06	693.56	3.117	1541.1	5.413	5.431	641.08
NEURALGCM ENS (50)	1.4°	12H	7.42	**606.84**	<u>2.756</u>	1374.2	**4.830**	<u>4.852</u>	—
ARCHESWEATHERGEN (MEAN)	1.5°	24H	0.99	<u>610.36</u>	**2.755**	<u>1373.7</u>	**4.830**	**4.848**	**567.15**
MOSAIC (MEAN; OURS)	1.5°	24H	**0.05**	624.08	2.778	**1358.1**	<u>4.880</u>	4.897	<u>576.44</u>

Table 1. Comparison of 1.5° resolution models on RMSE scores for key weather variables with 10-day lead-time against ERA5 data, 2020 test year. Best scores in **bold**, second best <u>underlined</u>. Results for additional variables (T2m, U10m, V10m) are provided in Table 9.
[†] Per-member-per-step inference time (seconds) on a single NVIDIA H100. See Table 8 for full benchmarking details.

batch size 2 on 8× NVIDIA H100 GPUs in fp16 precision. We use Muon (Jordan et al., 2024) optimizer, which is, to our knowledge, the first application of Muon to MLWPs. The pretraining stage takes 250,000 steps, while the fine-tuning phase includes multiple stages with progressively longer autoregressive rollouts (1, 2, 4, 8, 12 steps) for 60k total steps. We use the pushforward trick (Bodnar et al., 2025) where only the final unrolling step is backpropagated through. During training, we generate ensembles of size 2.

Evaluation We follow the WeatherBench 2 protocol (Rasp et al., 2023): all forecasts are initialized at 00:00 and 12:00 UTC and regridded to 1.5° before computing metrics. We report latitude-weighted root mean square error (RMSE) of both the first member and the ensemble mean, as well as the fair CRPS (Rasp et al., 2023) of the ensemble (**RQ1**). Metrics are computed at each grid point, then globally area-weighted and averaged for each variable and pressure level separately. To evaluate spectral fidelity, we report spectra of global kinetic energy (**RQ2**). To evaluate ensemble reliability, we report the spread-to-skill ratio – the ratio of ensemble spread to the RMSE of the ensemble mean – where values close to 1 indicate well-calibrated forecasts (**RQ3**). To assess whether spectral fidelity translates into skillful extreme event forecasting, we conduct an ensemble case study of Hurricane Ian using storm-center tracking (Brightband Technologies, 2026) (**RQ4**).

RQ1: Forecasting skill comparison

1.5° benchmark Table 1 reports 10-day RMSE for all headline variables against ERA5 ground truth. MOSAIC achieves the best Q700 RMSE (1358.1) among all models and the second-best scores on U850 (4.880) and SP (576.44), closely trailing ArchesWeatherGen and NeuralGCM. On Z500, MOSAIC (624.08) narrows the gap to the stronger probabilistic baselines NeuralGCM (606.84) and ArchesWeatherGen (610.36), while substantially outperforming the deterministic models Stormer (665.88) and ArchesWeather-Mx4 (693.56). Notably, MOSAIC achieves this performance at a fraction of the cost being 20× faster than ArchesWeatherGen and over 150× faster than NeuralGCM. As Fig. 2(c) shows, MOSAIC matches the forecast skill of the best probabilistic models while being an order of magnitude faster.

0.25° benchmark MOSAIC achieves competitive performance against 0.25° SOTA models. Fig. 6 shows RMSE and CRPS results across three headline variables: 2-meter temperature (2m temp), geopotential at 500 hPa (Z500), and 10-meter U-component of wind (U10). At 10-day lead time, MOSAIC achieves strong RMSE performance: on Z500, MOSAIC outperforms IFS-ENS (610 vs 622 m^2/s^2), Aurora (610 vs 668), Pangu (610 vs 800), and GraphCast (610 vs 748); on U10, MOSAIC surpasses IFS-ENS (3.35 vs 3.39 m/s) and all deterministic ML baselines; on 2m temperature, MOSAIC matches IFS-ENS (1.99 K) while outperforming Aurora, Pangu, and GraphCast.

For probabilistic metrics, MOSAIC demonstrates competitive CRPS scores as well. At 240h, MOSAIC achieves Z500 CRPS of 258, close to GenCast (254) and IFS-ENS (261), though FGN achieves the lowest (242). On U10, MOSAIC (1.58) outperforms IFS-ENS (1.61) and approaches GenCast (1.56). These results demonstrate that block-sparse attention operating at native resolution can extract sufficient information from 1.5° data to compete with models trained on 6× finer grids. Tables 10 and 11 provide RMSE at 24 h and 240 h for a broader set of headline variables; RMSE, CRPS, and spread-to-skill curves for the remaining variables (V10, MSL, U850, V850, Q700) are shown in Appendix B.1. Qualitative unrolling trajectories in Appendix B further illustrate MOSAIC's ability to maintain forecast quality over extended lead times.

RQ2: Spectral fidelity preservation

Spectral degradation is not specific to fine grids: it occurs at any resolution when the architecture or training procedure discards information the grid can represent. MOSAIC

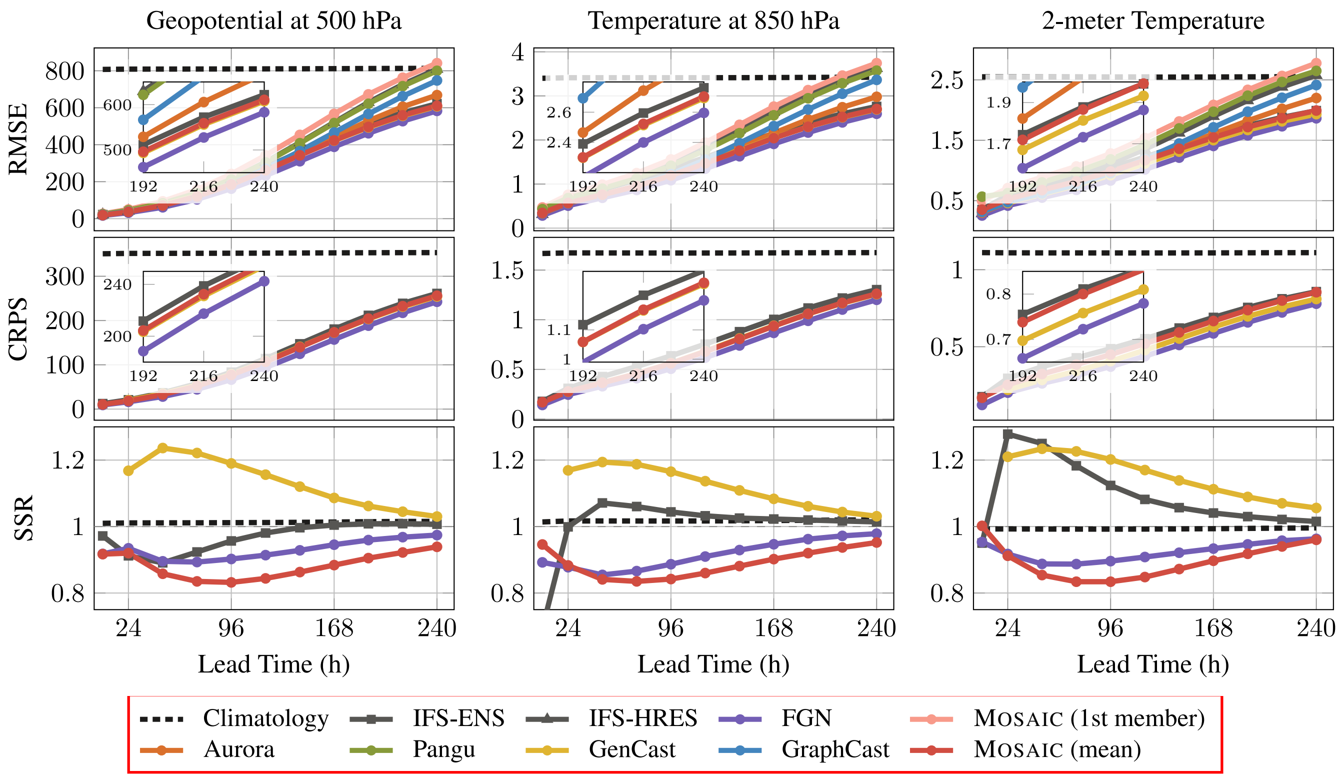

Figure 6. Forecast skill evaluated against IFS HRES Analysis following the WeatherBench 2 protocol (Rasp et al., 2023) on 2022 test year. Top row: RMSE; middle row: CRPS; bottom row: SSR.

operates at 1.5°, limiting our analysis to wavelengths above the Nyquist limit (∼333 km). Fig. 2(a) compares spectral power ratios of 10-meter wind speed for all ERA5-trained models at 1.5°. Deterministic models show clear spectral suppression at wavelengths the grid can resolve (Stormer energy ratio: 0.26; ArchesWeather-Mx4: 0.74), while individual ensemble members of probabilistic models track ERA5 faithfully (MOSAIC: 1.00; ARCHESGEN: 1.03). Ensemble means exhibit the expected suppression at higher frequencies due to smoothing from averaging (MOSAIC: 0.30; ARCHESGEN: 0.23). Fig. 2(b) extends this analysis to 0.25° models, where the same pattern holds: probabilistic members preserve spectral power while deterministic models diverge at fine scales. This validates our core hypothesis: by avoiding spatial compression and operating on native-resolution grids via block-sparse attention, MOSAIC learns the weather evolution function as faithfully as the data resolution permits.

RQ3: Ensemble calibration

Fig. 6 (bottom row) shows the spread-to-skill ratio across lead times. Well-calibrated ensembles should exhibit ratios close to 1, with values below 1 indicating overconfidence and values above 1 indicating underconfidence. MOSAIC achieves ratios ranging from 0.83 at medium lead times to 0.95 at 240h, closely matching FGN (0.89–0.97), while GenCast shows persistent underconfidence (1.03–1.24). This demonstrates that noise injection in SwiGLU gates produces well-calibrated uncertainty estimates on par with state-of-the-art probabilistic MLWPs.

RQ4: Extreme event forecasting: Hurricane Ian

To evaluate MOSAIC's ability to forecast extreme weather events, we conduct a case study of Hurricane Ian – a Category 4 hurricane that made landfall near Fort Myers, Florida, on September 28, 2022, with sustained winds of 155 mph. We generate 48-member ensembles from three initialization times (Sep 23, 25, and 27, 2022, all at 12Z) and track each member's storm center by locating the minimum mean sea level pressure (MSLP) within a search radius guided by IBTrACS best-track positions (Knapp et al., 2010), following the methodology of ExtremeWeatherBench (Brightband Technologies, 2026).

Fig. 7 shows the resulting ensemble track forecasts, with wind field evolution shown in Fig. 19. At 7-day lead time (Sep 23 init), the ensemble captures the general northward trajectory through the Caribbean, with multiple members already correctly tracking the recurvature over Cuba and landfall on the Florida Gulf Coast. At 5-day lead (Sep 25 init), the ensemble spread narrows and the ensemble mean closely follows the observed track through recurvature and approach toward landfall. At 3-day lead (Sep 27 init), the

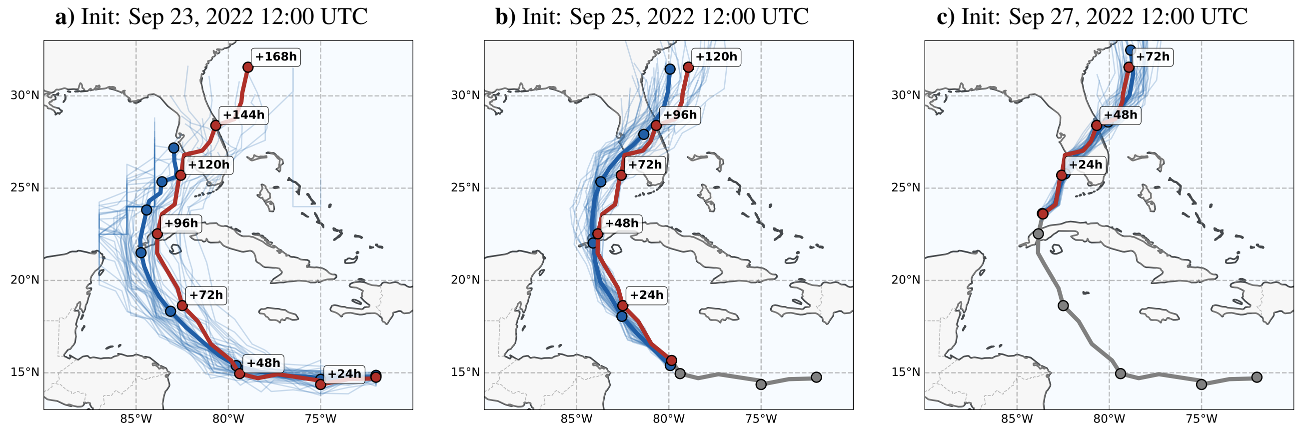

Figure 7. Hurricane Ian ensemble track forecasts from three initialization times: Sep 23 (7-day lead), Sep 25 (5-day lead), and Sep 27 (3-day lead), all in 2022. Each panel shows 48 ensemble member tracks (light blue), the ensemble mean (dark blue), and the observed track from HRES-fc0 (black). Markers indicate 24 h intervals. Storm centers are tracked via minimum MSLP guided by IBTrACS best-track positions (Knapp et al., 2010). Wind field evolution is shown in Fig. 19.

ensemble tightly brackets the observed track, with most members correctly identifying the landfall region. Despite operating at 1.5° resolution (∼166 km), which cannot resolve the inner-core structure of a tropical cyclone, MOSAIC manages to capture the hurricane's evolution. The progressive narrowing of ensemble spread with decreasing lead time demonstrates physically reasonable uncertainty quantification for an extreme event.

6. Conclusion

We introduced MOSAIC, a probabilistic weather forecasting model that addresses spectral degradation in MLWPs via (1) block-sparse attention, which processes native-resolution grids without compression at linear cost, and (2) learned functional perturbations, which produce ensemble members with realistic spectral variability. MOSAIC is able to match the forecasting skill of state-of-the-art models on both deterministic and probabilistic metrics,while being an order of magnitude faster. The spectral analysis reveals that individual ensemble members exhibit near-perfect spectral alignment across all resolved frequencies, and a case study of Hurricane Ian shows that this fidelity translates into skillful ensemble track forecasts of real-world extreme events.

Limitations and Future Work MOSAIC operates at 1.5° (∼166,km), which cannot resolve mesoscale phenomena such as tropical cyclone inner-core structure or individual severe thunderstorms. The linear cost and global receptive field of block-sparse attention make scaling to finer grids a natural next step: at 0.25° (∼700k tokens), where the architectural advantages over compression-based models should be most pronounced. We estimate that the training would require ∼20,000 H100 GPU-hours, which is comparable to existing 0.25° MLWPs. Beyond resolution scaling, MOSAIC's efficiency (12 s for a 24-member, 10-day forecast on a single GPU) opens the door to real-time ensemble nowcasting, rapid ensemble data assimilation cycles, and operational decision support. Finally, sparse attention over heterogeneous token sets could enable direct ingestion of sparse observations (radiosondes, aircraft reports, buoys) alongside gridded state tokens, potentially bypassing the costly analysis step that current MLWPs require.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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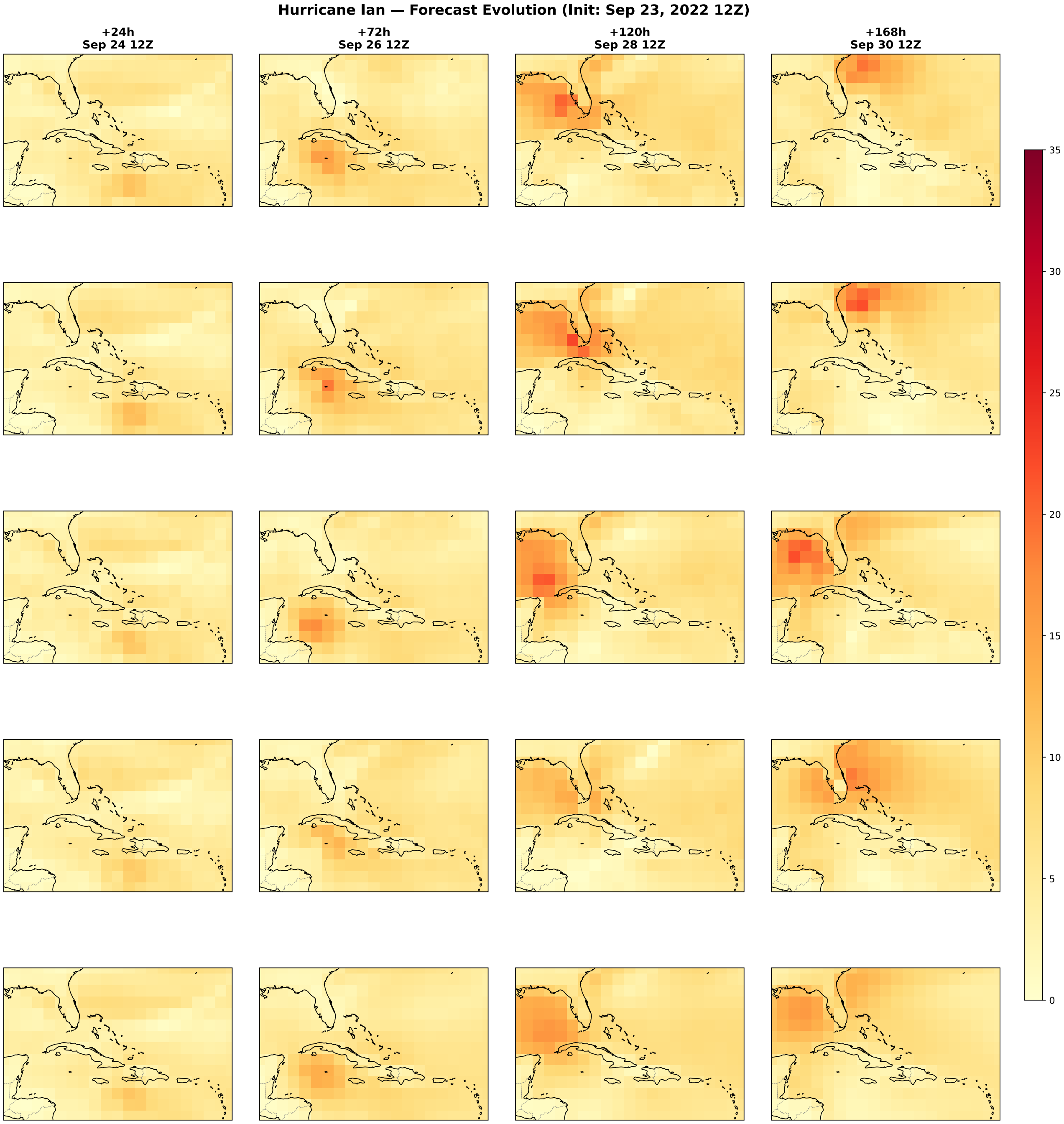

Figure 19. Hurricane Ian wind field evolution. Init: Sep 23, 2022 12Z. Columns show lead times (+24 h, +72 h, +120 h, +168 h). Rows: HRES-fc0 ground truth, three individual ensemble members, and ensemble mean (48 members). Variable: 10-meter wind speed (m/s). Region: Gulf of Mexico / Caribbean / southeastern US.

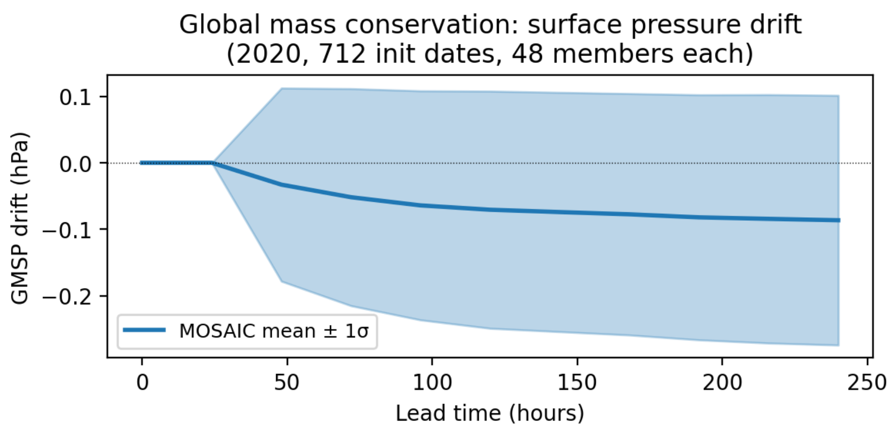

Figure 20. GMSP drift (mean $\pm$ 1σ) over lead time across 34,176 ten-day rollouts. The drift remains below 0.1 hPa throughout, indicating stable mass conservation.